Hoplynk Electrical Engineering Intern Take Home Assessment

As a next step in the process, we have a short electrical design exercise we would like you to complete. This exercise should take about 2–3 hours maximum.

As you know one of the projects we are actively working on is transitioning a deployable communications system from a prototype-style wired power architecture into a cleaner PCB-based power system.

The system includes:

- 4x lithium battery packs connected in parallel
- Starlink Mini
- Teltonika RUTM50
- Jetson Orin Nano Dev Kit
- Cudy GS105D 5 Port Gigabit Ethernet Switch
- charger / power supply input

Battery specs:

- 14.8V nominal voltage
- 6.4Ah per pack
- 16.8V charge cutoff voltage
- 11V discharge cutoff voltage
- 4 packs connected in parallel

The system should function as a UPS with battery backup and external power input.

Your task is to propose a cleaner PCB-based power architecture for the system.

Please think through things like:

- charger / power supply selection
- required voltages and currents
- connector selection
- power-path architecture
- protection circuitry
- battery monitoring
- load switching and power distribution
- what should vs should not be integrated onto the PCB
- cost, size, reliability, and manufacturability tradeoffs

Please send a short writeup that includes:

- a high-level power architecture / block diagram
- major assumptions and estimated current / power requirements
- your proposed protection / switching approach
- notes on tradeoffs, risks, and design decisions

Please also include some actual circuit design work in KiCad or another EDA tool. This does not need to be a complete PCB or full schematic, but we would like to see how you would begin implementing part of the system electrically.

We are mainly looking to see how you reason through the system, identify missing information, research components, make tradeoffs, and begin translating a messy real-world power architecture into an actual board-level design.

Please submit your results by Friday at 5 PM Pacific Time.